Objective 2: Linear Regression Analysis

Predicting football goal difference and team goals

Regression 2.1 predicts team1-minus-team2 goal difference; Regression 2.2 predicts individual team goals. Both use eight inputs for forecasts issued before each match day.

Data and Methodology

The provided data set of 2026 World Cup observations of teams and matches contains 104 different matches and 208 team–match observations that are included in the final CSVs. One signed goal difference and two team goal counts are given per match. According to FIFA (n.d.), the raw numbers are from the archive. Harmonized after harmonising team-names and accounting for a one-day date difference there are 36 dates differing from the archived martj42 (n.d.) cross-check snapshot, while all 104 scores agree. This checks for agreement between files supplied, not for independent authentication of their original downloads.

The evidence in the report is from Objective_2_Linear_Regression.ipynb and the final CSVs. This notebook is an update to the archived Part 3 and Part 4 models that switch to different predictors and splits.

The eight team1-minus-team2 variables are FIFA ranking points, World Cup matches played before 2026, host country status, membership of UEFA and CONMEBOL, previous goals conceded, previous shots, previuous expected goals (xG) and previous xG conceded. The uses for regression 2.2 include: team FIFA rank; opponent FIFA rank; team UEFA/CONMEBOL membership; opponent membership; team prior possession; team prior completed line breaks; team host-country status; and opponent prior xG conceded. The lower the rank the higher the ranking. There are no identically named shared columns, but four shared concepts: ranking strength, host status, confederation and xG.

Historical inputs are growing inputs from more restricted match dates, around the tournament mean at the time. No current, same day or future matches are included. Histories missing are ignored, and no history is available, it is replaced by a neutral centred value of zero. The selected CSV predictors have no missing or infinite values. Fitted separately within each fold, median imputation and standardisation are kept within each training pipeline. Observations will not be deleted. The release date for static ranking and their publication delay prior to the date of release are not confirmed and are therefore indicated in the collection notebook as 11 June 2026.

The hold-out period starts on 2 July 2026: 81 training matches and 23 test matches for 2.1; 162 training and 46 test matches for 2.2. Whole dates, and both teams, remain intact. Three growing windows are used for training evaluation which validate three following match days. Regularisation is tuned during each outer training window by two inner temporal splits. Lower RMSE in the lower mean is used to choose the family, and full-training temporal CV is used to choose the final parameters of the selected family. The earlier observed test-day results will update later historical inputs and fitted parameters will remain fixed. This is for the sequential forecasts and not for the whole tournament forecasted beforehand.

Results and Discussion

The selected models are compared in Table 1 for the same last 23 matches. R2 tests the mean of the test sample, while the baseline predicts the mean of the training sample. The temporal folds that CV RMSE averages are the same as MAE and RMSE are in goals.

Table 1. Verified model performance from the selected analysis.

Task	Model	CV RMSE	Test R^2	Test MAE	Test RMSE
2.1	ElasticNet (selected)	1.414	0.1747	1.158	1.503
2.1	OLS	1.436	0.1799	1.176	1.498
2.1	Mean baseline	1.922	-0.1152	1.461	1.747
2.2	Ridge (selected)	1.026	0.1304	0.920	1.226
2.2	OLS	1.028	0.1251	0.920	1.230
2.2	Mean baseline	1.221	-0.0004	1.063	1.315

Regression 2.1: predicting goal difference

ElasticNet achieved test MAE 1.158 and RMSE 1.503, compared with 1.461 and 1.747 for the baseline. The 14.0% RMSE reduction indicates useful signal in this late-tournament sample, although $R^2 = 0.175$ leaves most observed variation unexplained. Predictions overestimated the signed difference by 0.626 goals on average. Figure 1 shows that predicted differences do not consistently follow observed differences. The match-bootstrap 95% interval for MAE improvement spans −0.048 to 0.607 goals, so a reliable average benefit is not established by this small hold-out.

The largest standardised ElasticNet coefficient is FIFA points difference: +0.930 predicted goals per one training standard deviation, holding the other inputs fixed. Prior xG-conceded difference has coefficient −0.475 and World Cup experience +0.260. Prior xG difference is negative (−0.206), despite a positive football expectation for chance creation. This is a conditional association among correlated measures, rather than evidence that creating chances harms scoring. Penalisation sets prior goals-conceded and shots-difference coefficients to zero while preserving the eight explanatory inputs.

Regression 2.2: predicting team goals

Ridge achieved MAE 0.920 and RMSE 1.226, versus baseline values of 1.063 and 1.315. Its RMSE reduction is 6.8%; $R^2 = 0.130$ again indicates considerable unexplained variation. Mean residual is −0.025 goals, so average bias is small, but this hides errors for individual teams. In the supplied France–England match, England's six goals were predicted as 1.390, illustrating substantial underprediction of an extreme score. No selected-model test prediction was negative, although a linear model does not guarantee non-negative forecasts.

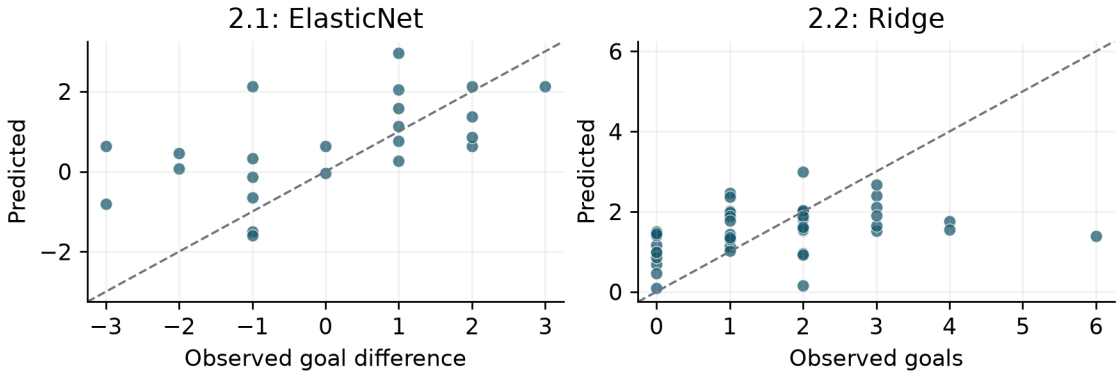


Figure 1. Locked-model predictions on the common hold-out. The dashed line denotes exact agreement; overlapping points may conceal repeated outcomes.

Opponent rank has a coefficient of +0.395 goals, which is what would be expected if they were more likely to score against a poorer opponent. The team's rank is correlated with itself at −0.257. Coefficients are conditional association with no causal implications for prior completed line breaks (+0.243) and host status (+0.163 per standard deviation). The 95% MAE-gain interval is from 0.035 to 0.245 goals, using match-bootstrap. This helps to achieve an improvement in the evaluated match, but the resampling of matches can't account for the dependence of repeated teams or uncertainty between tournaments.

Diagnostics and strength of evidence

Training OLS R^2 is 0.534 for 2.1 and 0.300 for 2.2, with adjusted R^2 of 0.482 and 0.264. Lower cautions against using training fit as evidence for prediction. Maximum VIFs of 3.168 and 4.671 are still below five, yet correlated inputs are still likely to impact on the stability of the coefficients. The Breusch–Pagan test output is $p = 0.367$ which does not demonstrate constant variance for 2.1; $p < 0.001$ does show nonconstant variance for 2.2. Eight and twenty training rows, respectively, are influence (or leverage) flags; all are saved. These diagnostics apply to OLS, rather than the penalised fits.

There are exploratory OLS match-cluster tests for points difference in 2.1 ($p = 0.002$) and for team and opponent rank in 2.2 ($p = 0.027$ and 0.009). They are not penalised-coefficient uncertain tests, but rather unadjusted tests after screening. Whereas team clustering handles pairs of rows, the clustering of rows for matches happens on rows.Team clustering handles paired rows, while match clustering handles rows. The held-out R^2 and baseline-relative improvement are both higher for regression 2.1. Lower raw RMSE for 2.2 does not make the task easier as the target distributions are different.

Limitations and Key Findings

The evidence suggests mild values above the training-mean baseline, especially the ranking strength. Both models account for little variation in scores. Short tournament histories, neutral zero priors, linear ranking proxies, and discrete outcomes and a late stage test sample limit interpretation and generalisation. There are no line-ups or injuries or changes in tactics.

There are no proof of the leak-free workflow in the cases of mutation checks or strict prior date construction. The results of the feature screening are not repeated within each outer fold, as they have already been explored in previous notebooks. The hold-out is even less suitable as a validation sample because of the lack of verified rank timestamps and download provenance. However, new tournaments and snapshots from pre-match are required to make more robust forecasting statements. The results do not imply causality or suitability for betting.

Data: Objective_2_Linear_Regression.ipynb, Sections 3–5 and 8–19, and final CSVs; FIFA (n.d.), World Cup 2026 statistics; International football results [Data set] (2022) by martj42.

AI assistance: Codex verified, drafted and formatted this report; group review, policy confirmation and a truthful declaration is pending.